

Introducción al uso de registros nacionales de salud para investigación

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Bienvenida



Bienvenidos al curso “Introducción al uso de registros nacionales de salud para investigación”. Este curso breve tiene como objetivo introducir los conceptos centrales del uso de registros de salud para investigación epidemiológica. El curso está enfocado en las enfermedades crónicas, especialmente en la diabetes pero los principios generales pueden aplicarse en otros contextos.

El curso se organiza por primera vez en 2026 en colaboración entre el [Grupo Crónicas](#), el [NCD Policy Lab](#) y [Steno Diabetes Center Aarhus](#).

Esta página web contiene todo el material del taller, incluyendo presentaciones y ejercicios. Está estructurado como un libro, con un “capítulo” para cada sesión. Haremos uso intensivo del internet durante el taller, donde las sesiones interactivas de programación en vivo siguen casi idénticamente los pasos en ésta página.

Datos clave

 10 de septiembre 2026

 9:00 - 12:00

 Universidad Peruana Cayetano Heredia, sede Miraflores, Lima, Perú

Público

El curso está diseñado como taller para investigadores jóvenes y estudiantes con conocimientos en análisis de datos interesados en conocer metodologías de análisis con registros nacionales de salud.

Preparación

Para agilizar las sesiones el día del taller, les pedimos que completen las [tareas preparatorias](#) antes de llegar a la clase.

Re-utilización y licencia

El taller se publica con una licencia [Licencia Internacional Pública de Atribución/Reconocimiento 4.0 de Creative Commons](#). Quiere decir que los materiales pueden ser utilizados, re-utilizados y adaptados, con tal que cualquier nueva versión o reproducción parcial o total contenga reconocimiento y atribución a éste material como fuente.

Como se construyó este taller

Los materiales de este taller fueron creados usando el formato [Quarto](#), [Git](#) y [GitHub](#) para la gestión del repositorio y Github Pages para crear y alojar la página web.



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1. Temario

Este taller dura 3 horas y consiste de las siguientes sesiones:

Consulta también el [horario](#)

- [Introducción al taller](#)
- [Potencial de los registros](#)
- [Partiendo de los datos crudos](#)
- [Un análisis sencillo](#)
- [Cierre](#)

1.1 Objetivos de aprendizaje

Hemos diseñado el taller para enseñarte lo siguiente:

1. Primer objetivo

2. ¿Este taller es para mí?

El curso está diseñado como taller para investigadores jóvenes y estudiantes con conocimientos en análisis de datos interesados en conocer metodologías de análisis con registros nacionales de salud.

Para ayudar a manejar las expectativas y para enfocar el desarrollo de los materiales, asumiremos cierta experiencia previa de los participantes de este taller. Esperamos que hayas estudiado, trabajado o tengas alguna experiencia con:

- Conceptos epidemiológicos básicos como prevalencia, determinantes, desenlaces, diseño de cohortes abiertas y cerradas, tiempo de seguimiento, censura, medidas de incidencia, medidas de riesgo, medidas de asociación, sesgos, confusión.
- Manejo de datos cuantitativos, conceptos básicos de la ciencia de datos como formatos *long* y *wide*, fusión de datos determinística y probabilística.
- Análisis de datos con programas estadísticos como R.

While we have these assumptions to help focus the content of the workshop, if you have an interest in the workshop but don't fit any of the assumptions, *you are still welcome to attend!* We welcome everyone until capacity has been reached.

In addition to the assumptions above, the workshop also has a fairly focused scope, which may also help you decide if this workshop is for you:

- TODO: List of concrete tasks we will do and not do.

3. Horario

Este taller contiene presentaciones, ejercicios participatorios con programación en vivo utilizando datos simulados, discusiones y trabajo en grupos.

3.1 10 de septiembre 2026

Horario	Módulo	Detalle	Expositor
09:00 - 09:10	Bienvenida	Presentación del docente y de las instituciones organizadoras.	
09:10 - 10:00	El potencial de los registros	Qué son los registros nacionales y poblacionales, y qué los hace tan valiosos. El modelo nórdico como ejemplo. Qué preguntas de investigación se vuelven posibles con estos datos que de otro modo serían muy difíciles. Fortalezas y limitaciones en lenguaje sencillo. Cierre del bloque contrastando con el contexto latinoamericano/peruano.	DW
10:00 - 10:15	Break		
10:15 - 11:00	De los datos crudos a una pregunta de investigación	Cómo se pasa de una base administrativa gigante a un conjunto de datos analizable. Ideas básicas: definir una población, elegir variables a partir de códigos, construir un seguimiento en el tiempo.	DW
11:00 - 11:50	Manos a la obra: un análisis sencillo	Un ejercicio guiado y simple con datos sintéticos o de ejemplo, donde todos siguen el mismo paso a paso: cargar los datos, mirar la estructura, y correr un análisis básico.	DW
11:50 - 12:00	Cierre. Ideas y discusión	Qué se necesitaría para hacer esto en nuestra	Todos

Horario	Módulo	Detalle	Expositor
		región/país y ronda de preguntas.	

4. Learning design

This workshop is designed using the science of learning. Learning is when new content gets incorporated into long-term storage in the brain in a way that it can be later retrieved. Some of the best ways to enable learning is to do “active learning” by engaging with the material, practicing retrieving what was just covered, repeatedly being exposed to the content, linking the new information with what you already know, and explaining it to someone else.

We use a variety of activities to target the different “pathways” of learning. A big part of learning is being engaged in the *process* of learning, not just learning the content itself. So to help you be engaged in your own learning, we want to involve you in the reasons *why* we do the things we do during this workshop.

Throughout the workshop, we use these practices to support learning.

4.1 Clear learning goals and a strong narrative

The learning goals and the narrative are tightly connected in this workshop. Learning is easier when you have a better idea of where you are going, where you are relative to the end goal, and what to expect along the way.

4.2 “Code-alongs” as the main activity

“Code-alongs” is when the teacher types on their computer that is on the projector and explains what it means as you also type along with them. The connection between using the hands and doing something tangible helps reinforce learning. It also helps the teachers to go slower as we have to both explain and type things out. This helps you learn better. “Less is more” works very well with learning.

4.3 Retrieval practice with hands-on exercises

Hands-on exercises reinforce what was just taught through retrieval practice and repetition. It gives you a chance to engage with the material and the space to try it out on your own. This helps you form a better sense of how to do it yourself.

4.4 Linking new information with existing knowledge with discussion activities

Discussion activities encourage linking what you already know and do in your work with the new information. Because humans are social beings, we also tend to learn better when there is some social aspect to it. For example, explaining something to someone else is one of the best ways to learn, as nicely expressed in the quote “while we teach, we learn” from the Roman philosopher Seneca.

4.5 Minimal use of lectures

Lectures, while extremely common, are forms of information sharing with the lowest rates of learning. In this workshop, we use reading tasks instead, so you can go at your pace, re-read things you may not understand, and take notes, all of which support learning. To help reinforce what you just read, the teacher will briefly summarise and emphasise key points after reading tasks, to repeat the main points and target another part of the brain (the auditory pathway).

4.6 Frequent breaks and variation in activities

Breaks are how your brain can recharge and refocus. The brain is only able to effectively focus on one thing at a time in 30-50 minute blocks of time. That’s why we mix together code-alongs, exercises, discussions,

and reading to give your brain some variation. Having an explicit break, even 5 minutes, is enough for the brain to better process and store what was covered.

4.7 Easily accessible material that is online, always

All the material is online on this website, so that it is easy for you to go back to after the workshop. This builds the repetition and retrieval practice part of learning. Our brains also get easily distracted, which is completely normal. So if you get distracted during the workshop, you can simply go to the material and get caught up with where we are. We also make sure to follow the material on the website closely, so that it is easier for you to follow along.

4.8 A safe and supportive environment

Mental safety in a supportive environment is critical for effective learning. It puts the brain into a more relaxed state that allows for optimal processing and ultimately learning. That's why we try to explicitly create that space, for example with the use of our Code of Conduct.

4.9 Using origami hats to ask for help

Asking for help can feel difficult, especially when you are learning and are unsure about something. To make it easier and feel safer, we use origami hats (or stickies) to show how everyone is doing. If you need help, you put up the "help" hat (usually orange) and a helper will come to you as soon as possible. If not, you can put up a "all good" hat (usually blue) to let us know you are following along. Since we often have several helpers, you usually won't wait long. This simple system lets you ask for help without interrupting the teacher or drawing attention, so you can keep following along while you wait. Our goal is to make getting help easier and more comfortable.

4.10 Being involved in the learning process

Now you know a bit more about the thoughts behind how we've designed this workshop to really try to support and enable your own learning! :tada: :grin:

II

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5. Tareas preparatorias antes del taller

Para agilizar el taller es importante que todos los participantes completen las siguientes tareas preparatorias:

1. Instala los programas R, RStudio, Quarto, y Git en sus versiones mas actuales de acuerdo con [?@sec-installing-programs](#).
2. Instala los paquetes R necesarios.
3. Lee el temario (Chapter 1.).

Cada sección contiene detalles sobre cómo completar cada tarea.

Tip

Puedes pasar a la siguiente sección haciendo click en la flechita en la parte inferior de cada página :

6. Instalación de los programas

Para poder participar en el taller, es importante que tengas los siguientes programas instalados en tu computadora. Es posible que ya hayas instalado algunos de ellos en el pasado. En ese caso por favor verifica que tengas instalada una versión mas reciente que la indicada en las instrucciones. Si no tendrás que actualizar las instalaciones.

6.1 Windows

1. **R**: Cualquier versión mas reciente que 4.6.0. Si has utilizado R anteriormente, puedes verificar tu versión actual dando éste comando en la consola: `R.version.string`.
2. **RStudio**: Cualquier versión mas reciente que 2026.08.2. Si has instalado RStudio anteriormente, puedes verificar la versión actual yendo al menu “Help -> About RStudio”.
3. **Git**: Escoge el enlace “Click here to download”. Usaremos Git durante algunas partes del taller. Al instalar Git, te pedirá seleccionar un “Text Editor” y aunque no usaremos esa funcion en el taller es necesario proporcionar el dato para la instalación de Git, así que lo mas fácil es escoger Notepad.

6.2 MacOS

1. **R**: Cualquier versión mas reciente que 4.6.0. Si has utilizado R anteriormente, puedes verificar tu versión actual dando éste comando en la consola: `R.version.string`. Si usas [Homebrew](#) (recomendado en general pero), la instalación de R is muy fácil. Solo hay que dar el siguiente comando en el Terminal de tu computadora:

```
brew install --cask r
```

2. **RStudio**: Cualquier versión mas reciente que 2026.08.2. Si has instalado RStudio anteriormente, puedes verificar la versión actual yendo al menu “Help -> About RStudio”. Con Homebrew:

```
brew install --cask rstudio
```

3. **Git**: Usaremos Git durante algunas partes del taller. MacOS viene con Git pre-instalado. Para instalarlo con Homebrew:

```
brew install git
```

6.3 Linux (Ubuntu)

1. **R**: Cualquier versión mas reciente que 4.6.0. Si has utilizado R anteriormente, puedes verificar tu versión actual dando éste comando en la consola: `R.version.string`.

```
sudo apt -y install r-base
```

2. **RStudio**: Cualquier versión mas reciente que 2026.08.2. Si has instalado RStudio anteriormente, puedes verificar la versión actual yendo al menu “Help -> About RStudio”.
3. **Git**: Usaremos Git durante algunas partes del taller.

```
sudo apt install git
```

Todos estos programas son necesarios para el taller, Git incluido. Git es usado para la gestión formal de versiones de archivos. Lo usamos porque tiene mucha documentación muy buena. Puedes aprender mas en éste libro en línea [Happy Git with R](#). Especialmente la sección “Why Git” explica bien por qué lo usamos. Usuarios de Windows a veces encuentran dificultades durante la instalación de Git. La siguiente guía ofrece pautas y ayuda [Installing Git for Windows](#).

7. Instalación de los paquetes R necesarios

We will be using specific R packages for the workshop, so you will need to install them. To make it easier, we've created a custom R package called `r3` that will install all the necessary packages for you for this workshop. Run these two functions in the Console by copying and pasting them to install our custom `r3` package as well as the `{tidyverse}` package.

```
install.packages("r3", repos = c("https://rostoools.r-universe.dev", "https://cloud.r-project.org"))
install.packages("tidyverse")
```

💡 Tip

Installing packages is tricky, especially on some work laptops, and doesn't always work as expected. Here are some other methods to try if you have trouble installing the packages with the code above.

For Windows users, you may have to install [RTools](#) to use either the `{pak}` or `{remotes}` method. The version of RTools you install must match the version of R that you have installed. For example, if you have R version 4.4, you need to install RTools 4.4.

First, try installing the `r3` package directly from GitHub with `{pak}`:

```
install.packages("pak")
pak::pak("rostoools/r3")
```

If that doesn't work, try with `{remotes}`:

```
install.packages("remotes")
remotes::install_github("rostoools/r3")
```

If you encounter any errors with either of these functions, try to restart R by going to “Sessions -> Restart R” and re-run them again. If that also doesn't work, try to complete the other tasks as best you can, complete the survey, and let us know you have a problem in the survey. We'll email you at least a week before the workshop starts to try to help you get it working.

💡 Tip

When you see a command like `something::something()`, for example with `pak::pak()`, you would “read” this as:

R, can you please use the `pak()` function from the `{pak}` package.

The normal way of doing this would be to load the package with `library(pak)` and then running the command `(pak())`. But by using the `::`, we tell R to directly use a function from a package, without needing to load the package and all of its other functions too. We use this trick because we only want to use the `pak()` command from the `{pak}` package and not have to load all the other functions as well. In this workshop we will be using `::` often.

8. Code of conduct

Learning happens when we feel safe and supported. Since we want you to learn during this workshop, we want you to feel safe and supported. We also want all learners to have an enjoyable and fulfilling experience while participating in this workshop. So we expect all learners to show respect, kindness, and courtesy to other learners throughout the whole workshop and to help create a positive and safe learning environment.

While most people and (sub-)cultures have an implicit, assumed set of practices and expectations for behaviour, we believe strongly in effective and clear communication. This includes making what is normally implicit expectations to instead be explicit expectations.

Teachers and helpers will uphold and embody this code in all spaces and activities of the workshop, and we expect you as the learner to do the same. In fact, by participating in this workshop, you indicate that you agree to follow these explicit expectations. We have this code of conduct to communicate to you that we really value and want to nurture a space that encourages and enhances learning. Otherwise, why would we have a workshop if you can't or aren't able to effectively learn anything from it! :zany_face:

8.1 Expected behavior

We want a welcoming, supportive, safe, and fun environment not only for ourselves as teachers and helpers, but also for all learners, regardless of background and identity. This means that we do not tolerate behaviour that is disrespectful to any of our teachers, helpers, or learners (see below for unacceptable behaviour).

We encourage the following kinds of behaviours in all interactions during this workshop that will foster a positive and professional learning environment:

- Show kindness, courtesy, and respect towards other workshop members.
- All interactions should be professional regardless of platform: either online or in-person.
- Be aware of your words by using welcoming and inclusive language. Words matter, a lot!
- Be respectful to different viewpoints and experiences.
- Gracefully accept constructive criticism.
- Focus on what is best for everyone in the workshop.

8.2 Unacceptable behavior

While we assume you will not do these behaviours, we want to clearly communicate what unacceptable behaviour is. It is any behaviour, as actions or words, that excludes, intimidates, or causes discomfort to others. We do not tolerate discrimination or harassment based on characteristics that include, though are not limited to, gender identity and expression, sexual orientation, disability, physical appearance, body size, citizenship, nationality, ethnic or social origin, pregnancy, familial status, veteran status, genetic information, religion or belief (or lack thereof), membership of a national minority, property, age, education, socio-economic status, technical choices, and experience level. We of course also don't tolerate verbally or physically violent behaviour towards others.

Examples of unacceptable behavior includes:

- Written or verbal comments that exclude people on the basis of membership of any specific group.
- The display of sexual or violent images.
- Unwelcome sexual attention, advances, and/or physical contact.
- Sustained disruption of workshop sessions or communications.
- Insults or put downs.
- Sexist, racist, homophobic, transphobic, ableist, or exclusionary jokes.

8.3 Consequences of unacceptable behavior

Learners who are asked, either in person or online, to stop any inappropriate behavior are expected to comply immediately. If a learner engages in behaviour that violates this code of conduct, the organizers, teachers, or helpers may take any action they deem appropriate, including warning the offender or removing them from the workshop.

If you feel in any way unsafe or mistreated, you may contact us privately—all communication is confidential.

Thank you for helping make this a welcoming, friendly workshop for all :smile:

Note

Acknowledgements: Large portions of this code of conduct were taken and modified from other codes of conduct, specifically [Software Carpentry](#) and [UofTCoders](#).

III

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9. Introducción al taller

Diapositivas de la introducción general en línea

i Nota para el profesor

Las diapositivas pueden contener notas que se pueden visualizar con la 'S' en el teclado

9.1 Tema de esta presentación

Esta presentación da una introducción general a los conceptos centrales del trabajo con registros sanitarios. Se dicta como primera sesión o como presentación pre-curso (en línea).

9.1.1 Objetivos de aprendizaje

Después de esta presentación los estudiantes tendrán los siguientes conocimientos:

- Qué es un registro sanitario; con cuales componentes debe contar
- A qué tipo de preguntas puede responder un registro
- Cuales son los tipos de usuarios de los datos de un registro sanitario
- A qué niveles existen datos / registros sanitarios y cuales son las necesidades de información a cada nivel

10. El potencial de los registros

En esta sesión repasaremos rápidamente algunas de las definiciones

10.1 Objetivos de aprendizaje

El objetivo de esta sesión es que el estudiante conozca y pueda explicar:

- Qué son los registros sanitarios poblacionales, con enfoque en enfermedades crónicas
- Cómo se construyen y mantienen
- Qué los hace tan valiosos
- Cual es la experiencia de los países nórdicos y de Brasil

Tomando el modelo nórdico como ejemplo discutiremos qué tipo de preguntas de investigación se vuelven posibles con estos datos que de otro modo serían muy difíciles. Discutiremos fortalezas, limitaciones y retos. Cerraremos el bloque contrastando con el contexto latinoamericano/peruano.

10.2 TODO: Section

i Reading task: ~4 minutes

TODO: Reading text

10.3 :technologist: Exercise: Title

Time: ~NUM minutes.

Text

10.4 :speech_balloon: Discussion activity: Title

Time: ~Num minutes.

Text

10.5 :book: Reading task: Title

Time: ~Num minutes.

Text

10.6 Summary

- TODO: List of summary items

10.7 Survey

Feedback survey! :tada:

11. Partiendo de datos crudos

11.1 Objetivos de aprendizaje

En esta sesión vamos a investigar la estructura de las bases de datos del registro de diabetes de Dinamarca, basándonos en datos simulados. Esto quiere decir que todos los datos que veremos son generados aleatóreamente y no pertenecen a ninguna persona real.

Usaremos el paquete “osdc” (Open Source Diabetes Classifier). Antes de iniciar esta sesión debes haber completado las tareas pre-taller.

11.2 Datos crudos

11.3 Datos analizables

11.4 Running Code

When you click the **Render** button a document will be generated that includes both content and the output of embedded code. You can embed code like this:

```
1 + 1
```

```
[1] 2
```

You can add options to executable code like this

```
[1] 4
```

The `echo: false` option disables the printing of code (only output is displayed).

12. TODO: Lesson title here

12.1 Learning objectives

The learning objectives for this session are:

1. Objective one

12.2 TODO: Section

i Reading task: ~4 minutes

TODO: Reading text

12.3 :technologist: Exercise: Title

Time: ~NUM minutes.

Text

12.4 :speech_balloon: Discussion activity: Title

Time: ~Num minutes.

Text

12.5 :book: Reading task: Title

Time: ~Num minutes.

Text

12.6 Summary

- TODO: List of summary items

12.7 Survey

Feedback survey! :tada:

13. What next

Slides

Teacher note

The slides contain speaking notes that you can view by pressing 'S' on the keyboard.

Appendices

A. For teachers

A more general guide to teaching or helping is found on the [Guides](#) website. This page has information relevant specifically to this workshop.

- For created teams, assign one teacher or helper to each group to help them specifically during the final group project.

References

B. Contributing

Check out the [Contributing Guide](#) for more details on how to contribute. For guidelines on how to be a helper or teacher, check out the [For teachers](#) page. As always, make sure to adhere to the [Code of Conduct](#).

B.1 :bug: Issues and bugs

The easiest way to contribute is to report issues or bugs that you might find while using or reading the workshop material. You can do this by creating a new issue on our GitHub repository.

B.2 :pencil2: Adding or modifying content

To contribute to the workshop, you first need to install [uv](#) and [justfile](#). We use [uv](#) and [justfile](#) to manage our project, such as to run checks and build the workshop website. Both the [uv](#) and [justfile](#) websites have a more detailed guide on using [uv](#), but below are some simple instructions to get you started.

It's easiest to [install uv](#) first and then install [justfile](#) using [uv](#). Then, to install [justfile](#), run:

```
uv tool install rust-just
```

We keep all our development workflows in the [justfile](#), so you can explore it to see what commands are available. To see a list of commands available, run:

```
just
```

As you contribute, make sure your changes will pass our checks by opening a terminal so that the working directory is the root of this workshop's parent/root folder and running:

```
just run-all
```

When committing changes, please try to follow [Conventional Commits](#) as Git messages. Using this convention allows us to be able to automatically create a release based on the commit message by using [Cocogitto](#). If you don't use Conventional Commits when making a commit, we will revise the pull request title to follow that format. That's because we use squash merges when merging pull requests, so all other commits in the pull request will be squashed into one commit.

B.3 :file_folder: Support and infrastructure files

- [.copier-answers.yml](#): Contains the answers you gave when copying the project from the template. **You should not modify this file directly.**
- [.gitignore](#): This ignore file tells Git which files to not track. Unless you know what you are doing, it's best to not touch this file.
- [_quarto.yml](#): Quarto configuration file for the website, including settings for the website, such as the theme, navigation, and other options.
- [_metadata.yml](#): Quarto metadata file for the website, including information about the project, such as the titles and GitHub names.
- [.pre-commit-config.yaml](#): [Pre-commit](#) configuration file for managing and running checks before each commit.
- [.config/](#): Contains configuration files for various tools used in the project, such as:
 - [typos.toml](#): [typos](#) spell checker configuration file.
 - [rmdl.toml](#) and [panache.toml](#): [rmdl](#) and [Panache](#) configuration file for formatting Markdown files in the project.
- [.zenodo.json](#): Structured citation metadata for your project when archived on [Zenodo](#). This is used to add the metadata to Zenodo when a GitHub release has been uploaded to Zenodo.
- [justfile](#): [just](#) configuration file for scripting project tasks.

- `.editorconfig`: Editor configuration file for [EditorConfig](#) to maintain consistent coding styles across different editors and IDEs.
- `CONTRIBUTING.md`: Guidelines for contributing to the project.
- `.github/`: Contains GitHub-specific files, such as issue and pull request templates, workflows, [dependabot](#) configuration, pull request templates, and a [CODEOWNERS](#) file.

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